8.1 ALTERNATOR - INTRODUCTION

The machine which generates alternating current is called as alternator or synchronous generator.

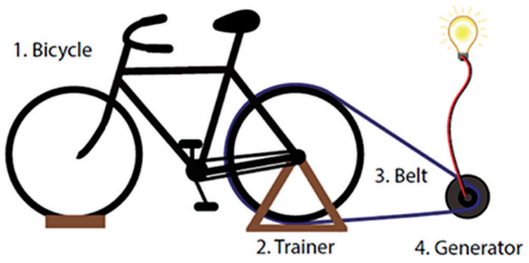


Fig 8.1 *Principle of AC Generation*

In Power Supply System alternating current is supplied to a greater extent that direct current supplies, because of the following reasons.

1. AC power can be generated in bulk quantity without much difficulty
2. AC requirements are cheaper in cost
3. AC voltage can be step up or step down by using transformers
4. AC can be easily converted into DC

8.1.1 Requirement of Alternator

For the generation of AC emf by the alternator the following basic systems are required:

1. Magnetic field system to produce the magnetic field.
2. Armature system which house the conductor on which emf is to be induced.
3. A prime mover is required which gives necessary rotational power for the generation of emf in the alternator.

8.1.2 Types of Alternator

According to the position of armature and field, alternators are classified into two types

a) Stationary field and rotating armature type.
b) Stationary armature and rotating field type.

a) **Stationary field and rotating armature type**

It is commonly used in small size generators and for the lower voltages. The DC supply given to the field windings and the alternating current generated is delivered to the slip ring. The mechanical construction of the revolving armature alternator is similar to that of the DC generator except that there is no commutator.

b) **Stationary armature and rotating field type**

Alternator with the stationary armature and revolving field type is used in the generation of high voltages.

The main reason for rotating armature type is the difficulty of using sliding contact brushes on slip rings at high voltages. With a stationary armature, the power from the generator is delivered through copper to copper connections firmly bolted together. The revolving fields are supplied with DC normally at 110V, through a pair of slip rings.

Advantages

1. It is easy to insulate the armature winding, because they are placed in the stator.
2. Output Current can be easily collected and easily supplied to the load circuit.

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